

2. Perform the following tasks using git-bash

- Initialize a Git repository for a data science project
- Push it to GitHub to enable version control
- Create a python file name as program1.py and write a python program to check even odd number.
- Commit the code and push the changes and check the changes on remote repository(GitHub)
- Create a new branch named GIT-EXAM and create new file named as program2 and write a program to find the reverse of a string
- Commit the changes and push them and demonstrate the difference of changes in master/main and GIT-EXAM branch
- Create a pull request to merge the changes from GIT-EXAM branch to master/main branch

a. Open **Git Bash** and run:

```
mkdir data-science-project
```

```
cd data-science-project
```

```
git init
```

✅ **b. Push it to GitHub**

- Go to GitHub and create a new repository (e.g., data-science-project)
- Connect local repo to GitHub:

```
git remote add origin https://github.com/your-username/data-science-project.git
```

OR

```
git clone https://github.com/your-username/data-science-project.git
```

```
cd data-science-project
```

✅ **c. Create program1.py (Multiplication Table)**

```
touch program1.py
```

code.

```
# Program to check even or odd number
```

```
num = int(input("Enter a number: "))
```

```
if num % 2 == 0:
```

```
    print("Even Number")
```

```
else:
```

```
    print("Odd Number")
```

✅ **d. Commit and push changes**

```
git add program1.py
```

```
git commit -m "Added program1.py for even odd check"
```

```
git push origin main
```

✅ **e. Create new branch GIT-EXAM & program2**

```
git checkout -b GIT-EXAM
```

```
touch program2.py
```

code.

```
# Program to reverse a string
```

```
string = input("Enter a string: ")
```

```
reverse = string[::-1]
```

```
print("Reversed string:", reverse)
```

✅ **f. Commit, push & show difference**

```
git add program2.py
```

```
git commit -m "Added program2.py for string reverse"
```

```
git push origin GIT-EXAM
```

✅ **g. Create Pull Request (on GitHub)**

1. Go to GitHub repository ,Click Compare & Pull Request
2. Select:
 - Base: main
 - Compare: GIT-EXAM
3. Click Create Pull Request
4. Click Merge Pull Request

Step 10: Pull Latest Changes (Optional)

```
git checkout main
```

```
git pull origin main
```